🧠 **JavaScript Practice Questions assignment**

✅ **Part 1: Conditions (if, if-else, else-if, ternary) – 10 Questions**

1. Check if a number is positive

Write a program that takes a number and checks if it is **greater than 0** using an if statement.

```
let num = 10;
if (num > 0) {
  console.log("The number is positive.");
}
```

---

2. Check if a number is even or odd

Use the modulus operator (%) and if-else to check whether a number is **even (divisible by 2)** or **odd**.

```
let num = 7;
if (num % 2 === 0) {
  console.log("Even");
} else {
  console.log("Odd");
}
```

---

3. Find the greater of two numbers

Accept two numbers. Use if-else to compare and print the **larger one**.

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4. Check if a person is eligible to vote

A person can vote if their age is **18 or above**. Write a program that checks eligibility based on the age.

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5. Check if a number is divisible by 5 and 11

A number is divisible by both if it gives a remainder 0 for **both 5 and 11**.

```
if (num % 5 === 0 && num % 11 === 0) {
  console.log("Divisible by 5 and 11");
```

```
} else {
  console.log("Not divisible by 5 and 11");
}
```

---

6. Assign grade using else-if

Given a score, assign grades as follows:

- A: 90 and above
- B: 80–89
- C: 70–79
- D: 60–69
- F: Below 60

Use else if ladder.

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7. Check if a character is a vowel or consonant

Input a single alphabet and check if it is one of a, e, i, o, u (vowel). Otherwise, it's a consonant.

---

8. Use ternary to check if number > 100

Use a **ternary operator** to print "Big" if number > 100, otherwise "Small".

```
let result = num > 100 ? "Big" : "Small";
```

---

9. Find the largest among 3 numbers using nested if

Accept 3 numbers and use if inside another if to compare and find the **largest**.

---

10. Check divisibility by 3 using ternary

Use ternary operator to check if a number is divisible by 3 and print the result.

🔁 **Part 2: Loops with Conditions (20 Questions)**

11. Print numbers 1 to 10

Use a for loop to print numbers from **1 to 10**.

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12. Print even numbers 1 to 50

Use loop + if to check if each number is divisible by 2, then print.

---

13. Print numbers divisible by 3 from 1 to 100

Loop through numbers 1–100 and print only if divisible by 3.

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14. Count numbers divisible by 7 between 1–100

Use a counter to track how many numbers in 1–100 are divisible by 7.

---

15. Sum of odd numbers from 1 to 50

Use loop + if (num % 2 !== 0) to add only odd numbers.

---

16. Print square of numbers 1 to 10

Use for loop and print each number's square using Math.pow() or n * n.

---

17. Print only positive numbers from an array

Given an array, loop through it and use if to print only numbers > 0.

let arr = [2, -5, 6, -1, 3, -9];

---

18. Count numbers > 10 in array

Count how many elements in the array are greater than 10.

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19. FizzBuzz (classic problem)

For numbers 1–30:

- Print Fizz if divisible by 3
- Print Buzz if divisible by 5
- Print FizzBuzz if divisible by both

---

20. Check if a number is prime

A prime number is only divisible by 1 and itself. Loop from 2 to n - 1 and check if divisible by any number.

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21. Sum of digits of a number

Given a number like 123, break it into digits and sum them: 1 + 2 + 3 = 6.

---

22. Reverse a number

Use loop to extract digits from the end and build the reverse. (e.g., 123 → 321)

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23. Factorial of a number

Factorial of n is n * (n-1) * ... * 1. Use a loop to calculate factorial.

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24. Print first 10 multiples of a number

Input a number and print its first 10 multiples using a loop.

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25. Count vowels in a string

Loop through a string and count how many characters are vowels (a, e, i, o, u).

---

26. Print even numbers from array

Given an array, print only the even values.

---

27. Find largest number in array

Loop through the array, compare and track the largest number.

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28. Print “Pass” or “Fail” using ternary

For each mark in an array, use ternary:

mark >= 40 ? "Pass" : "Fail"

---

29. Count negative numbers in an array

Use loop + if (num < 0) to count how many elements are negative.

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30. Print numbers divisible by 3 but not 9

Loop through array or range. Use:

if (num % 3 === 0 && num % 9 !== 0)